Ankur Barsode

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PhD candidate (Gravitational wave astronomy) at the International Centre for Theoretical Science, Bangalore, expected August 2026

MSc (Physics) at Birla Institute of Technology and Science Pilani, Rajasthan, 2021

Research Interests

Detecting gravitational lensing of gravitational waves (GWs), astrophysics and cosmology using GWs, Bayesian statistics and data analysis

Experience

- PhD Student, ICTS-TIFR, Bangalore (2021-present)
- Short-term Visitor, Albert Einstein Institute (AEI), Potsdam (2024)
- Short-term Visitor, UCLouvain, Belgium (2024)
- Long-term Research Student, National Center for Radio Astronomy, Pune (2021)
- Summer Research Student, National Center for Radio Astronomy, Pune (2020)

LIGO-Virgo-Kagra

- Data analyst: search for strongly lensed GWs (2022-present)
- Editorial Team member: search for lensed GWs in O4b (2025-present)

Honors and Awards

- International Travel Support from ANRF for presenting my work at the Summer School in Cosmology at International Centre for Theoretical Physics (ICTP), Trieste (2024)
- Prabhat Award by BITS Alumnus to the best outgoing research oriented student of physics (2021)
- KVPY by the Department of Science and Technology - Govt. of India to encourage research at the undergraduate level (2016)

Talks and Presentations

Invited Talks and Seminars

- "Detecting strongly lensed gravitational waves," IISc, India (2026)
- "GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures" LVK Webinar (2026)
- "PO2.0: Fast and efficient GW strong lensing search," AEI, Potsdam (2024)
- "PO2.0: Fast and efficient GW strong lensing search," UCLouvain (2024)

Contributed Talks and Posters

- "Constraints on compact DM from the non-observation of GW strong lensing," Summer School in Cosmology, ICTP, Trieste (2024)
- "Hybrid methods to simulate the cosmological HI field," Frontiers in Cosmology, Raman Research Institute, Bangalore (2023)

– "Constraints on compact DM from the non-observation of GW strong lensing," IAGRG, Indian Institute of Science Education and Research, Kolkata (2022)

Teaching and Mentoring

– Lecturer, GW-Open Data Workshop, ICTS-TIFR, Bangalore (2025-2026)
– Tutor, GW-Open Data Workshop, ICTS-TIFR, Bangalore (2024-2026)
– Tutor, Summer School on GW astronomy, ICTS-TIFR, Bangalore (2023-2025)
– Teaching Assistant, Tools of Astrophysics, ICTS-TIFR, Bangalore (2024)
– Mentoring a first-year PhD student in research basics (2025-present)
– Informally advised a Master's student (2025)

Service and Outreach

– Reviewer, LIGO-Virgo-KAGRA's Papers and Publications
– Organizer, GW-Open Data Workshop, ICTS-TIFR, Bangalore (2024-2025)
– Invented the The Gravitational Wave Chirp Bottle to take GW astronomy to households
– Organized and conducted public demonstrations as a member of Astro Club (ICTS-TIFR)
– Contributed to an exhibition on Climate Change (ICTS-TIFR)

Skills

Programming	Python, C++, JavaScript; git, large scale computing, pipelining
Analysis	gravitational wave data analysis, statistics, parameter estimation, machine learning
Visualization	Matplotlib, Processing/p5js, Blender
Communication	Scientific writing (LaTeX, Markdown), presentation (slides, posters, blackboard)
Miscellaneous	Arduino, workshop

Selected Publications

– Barsode, A., Maity, K. N., & Ajith, P. (2026). Lensing, not luck! Detection prospects of strongly lensed gravitational waves. ApJ, 998(2), 294.
– Barsode, A., Goyal, S., & Ajith, P. (2025). Fast and efficient Bayesian method to search for strongly lensed gravitational waves. ApJ, 980(2), 258.
– Barsode, A., Kapadia, S. J., & Ajith, P. (2024). Constraints on compact dark matter from the nonobservation of gravitational-wave strong lensing. ApJ, 975(1), 48.
– Barsode, A., & Choudhury, T. R. (2024). Efficient hybrid technique for generating sub-grid haloes in reionization simulations. JCAP, 2024(11), 036.

Full publication list available on Inspire-HEP, Google Scholar